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Development of an Arduino-Based RADAR System for Object Detection and Distance Measurement

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Introduction: Radio Detection and Ranging (RADAR) systems have gone through substantial progress since their inception in the 1930s. Previously, this system was very bulky and applied only to military services. Now, it is available in compact and commercially viable forms. This has become possible due to the progressions in integrated circuits (ICs) and monolithic microwave integrated circuits (MMICs) and applicable to several areas^{[1]-[4]}. This paper explores the development of a low-cost, Arduino-based radar system for short-range object detection and distance measurement.

Methodology: This system used an Arduino Uno R3, a sonar sensor (HCSR04), a servo motor, a buzzer, a 9V battery, and some connecting wires as shown in Fig. 1. Arduino is programmed with the C-like language in an Integrated Development Environment (IDE), and this code is then uploaded to the Arduino Uno Board^[5-6]. The system runs by the 9V battery. The proposed system detects objects within a 180° range, which is controlled by the servo motor. The Arduino Uno processes sensor data to calculate the distance and angle of detected objects, which are then visualized by the IDE. A buzzer is used to make sounds when objects come very closer.

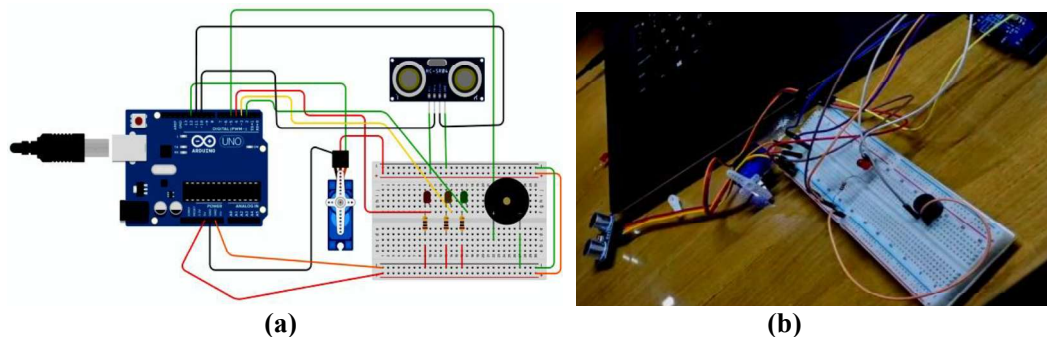


Fig. 1: (a) Schematic circuit diagram and (b) hardware connection with the laptop

Results: The developed RADAR system was tested for a series of situations to evaluate its performance. Successfully detected objects within a range of 2 cm to 400 cm, achieving a consistent measurement accuracy of ± 1 cm for objects within 200 cm, and maintained ± 3 cm accuracy at maximum range of 400 cm. However, the performance degradation was observed beyond 300 cm due to ultrasonic signal attenuation. The total cost of the system was BDT1650 (Bangladeshi Taka sixteen hundred fifty only).

The hardware system outputs are depicted in Fig. 2. It is observed that when there is an object then a red line appears but if there is no object all lines are green in color.

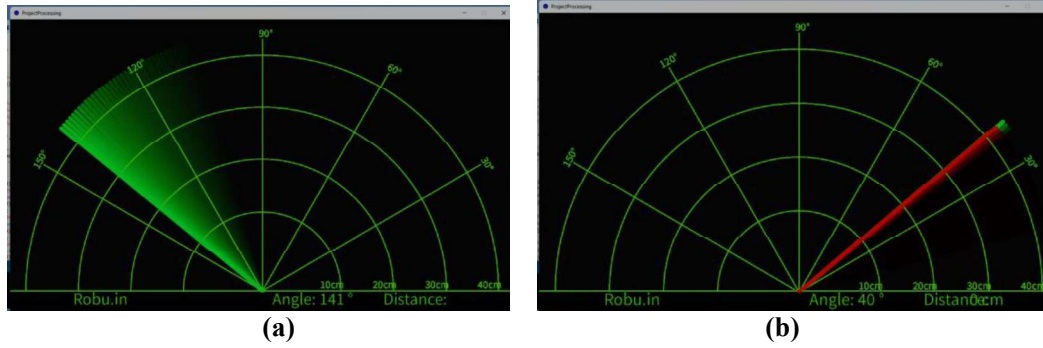


Fig. 2: Experimental outputs of the system: (a) no object in front of the sensor, (b) an object comes in front of the sensor at angle of 40°

Conclusion: The Arduino-based mini-RADAR system is developed in this work. This offers a cost-effective and scalable solution for short-range object detection and distance measurement. In future, multiple sensors may be integrated with it for extending the range of detection or improving the resolution with higher operating frequency. Potential application areas may be in autonomous vehicles and smart home systems.

Keywords: RADAR, Arduino Uno, Ultrasonic Sensor, Buzzer, Servo Motor,.

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